

7.2 Inverse Transforms and Transforms of Derivatives

The Laplace Transform takes a function $f(t)$ and transforms it into $F(s)$.

$$\mathcal{L}\{f(t)\} = F(s)$$

The inverse Laplace transform takes $F(s)$ as an input and returns $f(t)$ as the output.

$$\mathcal{L}^{-1}\{F(s)\} = f(t)$$

↓
inverse laplace
transform

$$\mathcal{L}\{1\} = 1/s \Rightarrow \mathcal{L}^{-1}\{1/s\} = 1$$

$$\mathcal{L}\{e^{3t}\} = \frac{1}{s-3} \Rightarrow \mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\} = e^{3t}$$

$$\mathcal{L}\{\cos(at)\} = \frac{s}{s^2+4} \Rightarrow \mathcal{L}^{-1}\left\{\frac{s}{s^2+4}\right\} = \cos(at)$$

$\mathcal{L}^{-1}$ is Linear

More specifically,

$$\mathcal{L}^{-1}\{\alpha F(s) + \beta G(s)\} = \alpha \mathcal{L}^{-1}\{F(s)\} + \beta \mathcal{L}^{-1}\{G(s)\}$$

For any constants α and β .

ex: ① Evaluate $\mathcal{L}^{-1}\left\{\frac{3s+6}{s^2+49}\right\}$

$$\frac{3s+6}{s^2+49} = 3\left(\frac{s}{s^2+49}\right) + \frac{6}{\textcircled{7}}\left(\frac{\textcircled{7}}{s^2+49}\right)$$

Note: We multiplied and divided by $\textcircled{7}$ to match the pattern $\frac{\omega}{s^2+\omega^2}$

Since: $\mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2+\omega^2}$

and

$$\mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{s^2+\omega^2}$$

From the
Table of
Laplace
Transforms

Then :

$$\mathcal{L}^{-1} \left\{ \frac{3s+6}{s^2+49} \right\} = 3 \mathcal{L}^{-1} \left\{ \frac{s}{s^2+49} \right\} + \frac{6}{7} \mathcal{L}^{-1} \left\{ \frac{7}{s^2+49} \right\}$$

$$= \boxed{3 \cos(7t) + \frac{6}{7} \sin(7t)}$$

ex: ② Evaluate $\mathcal{L}^{-1} \left\{ \frac{3s+4}{(s+1)(s+3)} \right\}$

We first do a Partial Fractions decomposition.

$$\frac{3s+4}{(s+1)(s+3)} = \frac{A}{s+1} + \frac{B}{s+3}$$

Various methods are available to find A and B, use any method that you feel comfortable with.

$$3s+4 = A(s+3) + B(s+1)$$

$$s = -1 \rightarrow -3+4 = A(2) + 0 \rightarrow A = 1/2$$

$$s = -3 \rightarrow -9+4 = 0 + B(-2) \rightarrow B = 5/2$$

$$\frac{3s+4}{(s+1)(s+3)} = \frac{1/2}{s+1} + \frac{5/2}{s+3}$$

$$\text{since: } \mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$$

$$\Rightarrow \mathcal{L}^{-1}\left\{\frac{3s+4}{(s+1)(s+3)}\right\} = \frac{1}{2}e^{-t} + \frac{5}{2}e^{-3t}$$

$$\underline{\text{Note:}} \quad s+1 = s - (-1)$$

and

$$s+3 = s - (-3)$$

ex: ③ Evaluate $\mathcal{L}^{-1} \left\{ \frac{s}{(s+1)(s^2+9)} \right\}$.

Let's use Partial Fractions (P.F.).

$$\frac{s}{(s+1)(s^2+9)} = \frac{A}{s+1} + \frac{Bs+C}{s^2+9}$$

$$s = A(s^2+9) + (Bs+C)(s+1)$$

$$s = -1 \rightarrow -1 = 10A + 0 \rightarrow A = -1/10$$

$$s = 0 \rightarrow 0 = 9A + C = -\frac{9}{10} + C \rightarrow C = 9/10$$

$$s = 1 \rightarrow 1 = 10A + 2B + 2C$$

$$1 = 10\left(-\frac{1}{10}\right) + 2B + 2\left(\frac{9}{10}\right) \rightarrow B = 1/10$$

$$\frac{s}{(s+1)(s^2+9)} = \frac{-1/10}{s+1} + \frac{\frac{1}{10}s + \frac{9}{10}}{s^2+9}$$

$$\frac{s}{(s+1)(s^2+9)} = -\frac{1}{10} \left(\frac{1}{s+1} \right) + \frac{1}{10} \left(\frac{s}{s^2+9} \right) + \frac{3}{10} \left(\frac{3}{s^2+9} \right)$$

$$\mathcal{L}^{-1} \left\{ \frac{s}{(s+1)(s^2+9)} \right\} = -\frac{1}{10} e^{-t} + \frac{1}{10} \cos(3t) + \frac{3}{10} \sin(3t)$$

ex: ④ Evaluate $\mathcal{L}^{-1} \left\{ \frac{3s-2}{s^3(s^2+4)} \right\}$.

Let's use Partial Fractions.

$$\frac{3s-2}{s^3(s^2+4)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{Ds+E}{s^2+4}$$

$$3s-2 = As^2(s^2+4) + Bs(s^2+4) + C(s^2+4) + (Ds+E)s^3$$

Let's expand and match coefficients.

$$3s - 2 = As^4 + 4As^2 + Bs^3 + 4Bs + Cs^2 + 4C + Ds^4 + Es^3$$

$$3s - 2 = (A+D)s^4 + (B+E)s^3 + (4A+C)s^2 + 4Bs + 4C$$

Note that: $3s - 2 = 0s^4 + 0s^3 + 0s^2 + 3s - 2$

$$\left\{ \begin{array}{l} A+D=0 \longrightarrow D=-A=-1/8 \\ B+E=0 \longrightarrow E=-B=-3/4 \\ 4A+C=0 \longrightarrow 4A-1/2=0 \longrightarrow A=1/8 \\ 4B=3 \longrightarrow B=3/4 \\ 4C=-2 \longrightarrow C=-1/2 \end{array} \right.$$

Let $F(s) = \frac{s}{(s+1)(s^2+4)}$

$$F(s) = \frac{1/8}{s} + \frac{3/4}{s^2} + \frac{(-1/2)}{s^2} + \frac{(-1/8)s - 3/4}{s^2+4}$$

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$

$$\Rightarrow \mathcal{L}^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{n!}$$

$$F(s) = \frac{1}{8}\left(\frac{1}{s}\right) + \frac{3}{4}\left(\frac{1}{s^2}\right) - \frac{1}{2}\left(\frac{1}{s^3}\right) - \frac{1}{8}\left(\frac{s}{s^2+4}\right) - \frac{3}{8}\left(\frac{2}{s^2+4}\right)$$

$$\mathcal{L}^{-1}\{F(s)\} = \frac{1}{8} + \frac{3}{4}t - \frac{1}{2}\left(\frac{t^2}{2!}\right) - \frac{1}{8}\cos(at) - \frac{3}{8}\sin(at)$$

$$\mathcal{L}^{-1}\{F(s)\} = \boxed{\frac{1}{8} + \frac{3t}{4} - \frac{t^2}{4} - \frac{1}{8}\cos(at) - \frac{3}{8}\sin(at)}$$

What is $\mathcal{L}\{y'(t)\}$?

By definition :

$$\mathcal{L}\{y'(t)\} = \int_0^{\infty} e^{-st} y'(t) dt$$

Let's use integration by parts with

$$\begin{cases} u = e^{-st} & dv = y'(t) dt \\ du = -s e^{-st} dt & v = y(t) \end{cases}$$

$$\mathcal{L}\{y'(t)\} = y(t)e^{-st} \Big|_0^{\infty} + s \int_0^{\infty} e^{-st} y(t) dt$$

$$= \lim_{t \rightarrow \infty} \frac{y(t)}{e^{st}} - y(0) + s \mathcal{L}\{y(t)\}$$

We assume that for sufficiently large $s > 0$,

$$\lim_{t \rightarrow \infty} \frac{y(t)}{e^{st}} = 0$$

$$\Rightarrow \mathcal{L}\{y'(t)\} = s \mathcal{L}\{y(t)\} - y(0)$$

or

$$\mathcal{L}\{y'(t)\} = s Y(s) - y(0)$$

Since $y''(t) = [y'(t)]'$

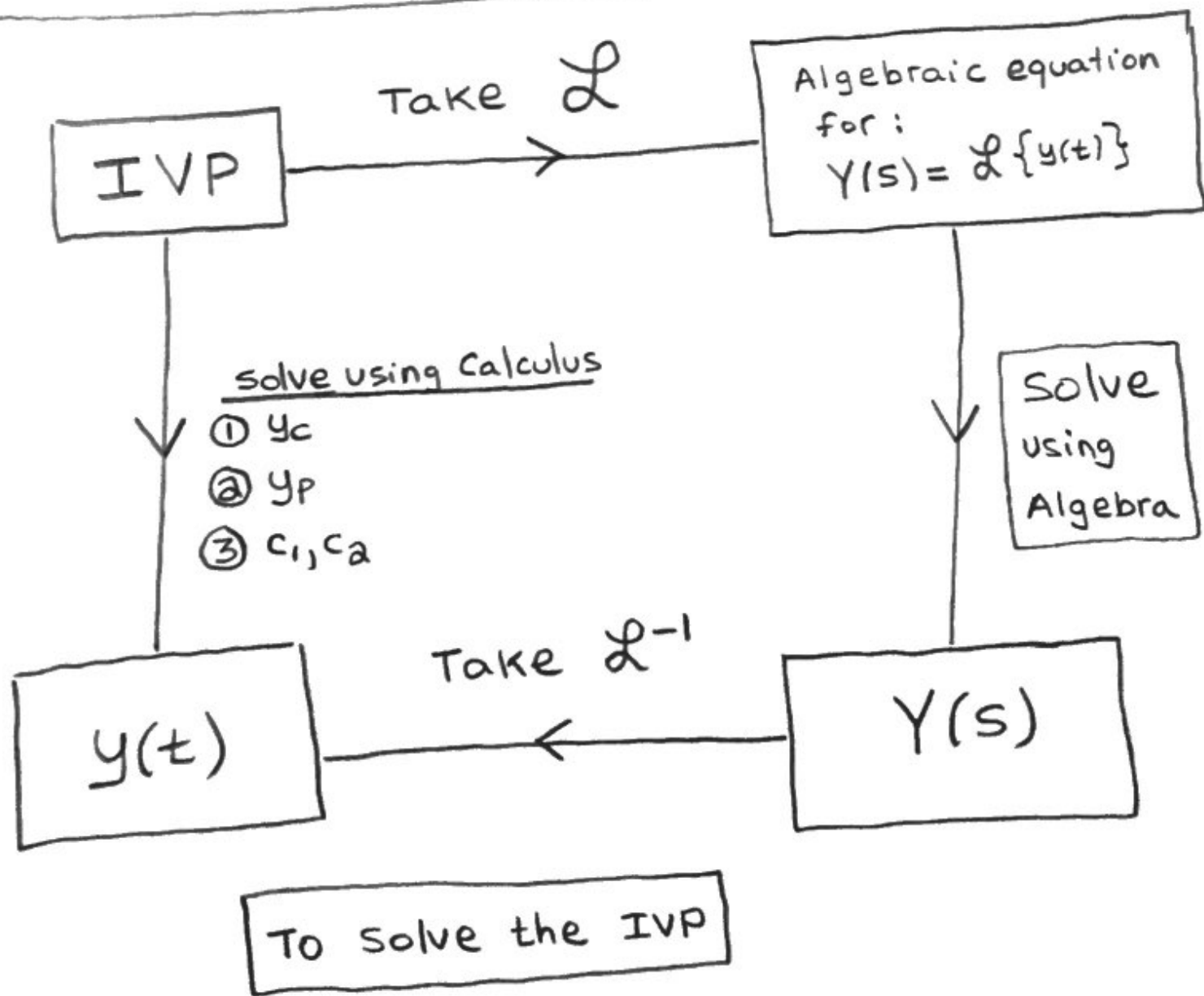
then:
$$\begin{aligned}\mathcal{L}\{y''(t)\} &= s \mathcal{L}\{y'(t)\} - y'(0) \\ &= s [s \mathcal{L}\{y(t)\} - y(0)] - y'(0) \\ &= s^2 \mathcal{L}\{y(t)\} - sy(0) - y'(0)\end{aligned}$$

or

$$\mathcal{L}\{y''(t)\} = s^2 Y(s) - sy(0) - y'(0)$$

Consider the IVP

$$\begin{cases} ay''(t) + by'(t) + cy(t) = f(t) \\ y(0) = y_0 \\ y'(0) = y_1 \end{cases}$$



ex: ⑤ Solve the IVP

$$\begin{cases} y''(t) + y'(t) - 6y(t) = 3e^{-t} \\ y(0) = 1 \\ y'(0) = 5 \end{cases}$$

Take Laplace transform on both sides of the DE.

$$\mathcal{L}\{y''(t)\} + \mathcal{L}\{y'(t)\} - 6\mathcal{L}\{y(t)\} = 3\mathcal{L}\{e^{-t}\}$$

$$\left[s^2 Y(s) - sy(0) - y'(0) \right] + \left[sY(s) - y(0) \right] - 6Y(s)$$

$$= 3 \left(\frac{1}{s+1} \right)$$

$$\Rightarrow (s^2 + s - 6)Y(s) - s - 5 - 1 = \frac{3}{s+1}$$

$$\Rightarrow Y(s) = \frac{s+6}{s^2+s-6} + \frac{3}{(s+1)(s^2+s-6)}$$

$$s^2 + s - 6 = (s+3)(s-2)$$

$$\Rightarrow Y(s) = \frac{(s+6)(s+1) + 3}{(s+3)(s-2)(s+1)}$$

Let's use P.F.

$$Y(s) = \frac{A}{s+3} + \frac{B}{s-2} + \frac{C}{s+1}$$

Solving for A, B, C gives:

$$A = -3/10, \quad B = 9/5, \quad C = -1/2$$

$$y(t) = \mathcal{L}^{-1}\{Y(s)\}$$

$$y(t) = -\frac{3}{10}e^{-3t} + \frac{9}{5}e^{2t} - \frac{1}{2}e^{-t}$$

ex: ⑥ Solve the IVP

$$\begin{cases} y'(t) + 3y(t) = \cos(2t) \\ y(0) = 2 \end{cases}$$

$$\mathcal{L}\{y'(t)\} + 3\mathcal{L}\{y(t)\} = \mathcal{L}\{\cos(2t)\}$$

$$[sY(s) - y(0)] + 3Y(s) = \frac{s}{s^2 + 4}$$

$$(s+3)Y(s) - 2 = \frac{s}{s^2 + 4}$$

$$\Rightarrow Y(s) = \frac{2}{s+3} + \boxed{\frac{s}{(s+3)(s^2+4)}}$$

We can
invert this
term.

This term requires
P.F.

P.F.

$$\frac{S}{(s+3)(s^2+4)} = \frac{A}{s+3} + \frac{Bs+C}{s^2+4}$$

$$S = A(s^2+4) + (Bs+C)(s+3)$$

$$s = -3 \rightarrow -3 = 13A + 0 \rightarrow A = -3/13$$

$$s = 0 \rightarrow 0 = 4A + 3C \rightarrow C = 4/13$$

$$s = 1 \rightarrow 1 = 5A + 4B + 4C \rightarrow B = 3/13$$

$$Y(s) = \frac{2}{s+3} + \frac{(-3/13)}{s+3} + \frac{(\frac{3}{13}s + \frac{4}{13})}{s^2+4}$$

$$Y(s) = \frac{23}{13} \left(\frac{1}{s+3} \right) + \frac{3}{13} \left(\frac{s}{s^2+4} \right) + \frac{2}{13} \left(\frac{2}{s^2+4} \right)$$

$$y(t) = \mathcal{L}^{-1}\{Y(s)\}$$

$$\Rightarrow y(t) = \frac{23}{13} e^{-3t} + \frac{3}{13} \cos(2t) + \frac{2}{13} \sin(2t)$$